PHYSICS

1) The main difference in the phenomenon of interference and diffraction is that :-

(1) diffraction is due to interaction of light from the same wavefront whereas interference is the interaction of waves from two isolated sources.

(2) diffraction is due to interaction of light from same wavefront, whereas the interference is the interaction of two waves derived from the same source.

(3) diffraction is due to interaction of waves derived from the same source, whereas the interference is the bending of light from the same wavefront.

(4) diffraction is caused by the reflected waves from a source whereas interference is caused due to refraction of waves from a source.

2) What should be the width of each slit to obtain 20 maxima of the double slit pattern with in the central maximum of the single slit pattern ? (Distance between both slit = 2 mm)

(1) 0.2 mm

(2) 2 mm

(3) 4 mm

(4) 0.6 mm

3) Polarising angle for water is 53°4'. If light is incident at this angle on the surface of water and reflected, the angle of refraction is :

(1) 53°4'

(2) 126°56'

(3) 36°56'

(4) 30°4'

4) Four polaroids are placed such that the optic axis of each is inclined at an angle of 30° with the optic axis of the preceding one. If unpolarised light of intensity I_0 falls on the first polaroid, the light transmitted from the fourth is :-

(1) I_0

(2) $I_0/16$

(3) $0.21\ I_0$

(4) $\frac{I_0}{2}$

5) Determine the angular separation between central maximum and first order maxima of the diffraction pattern due to a single slit of width 0.25 mm when light of wavelength 5890 Å is incident on it:-

(1) 3.534×10^{-3} rad

(2) 2.35×10^{-3} rad

(3) 1.47×10^{-4} rad

(4) 15.8×10^{-3} rad

6) Find the half angular width of the central bright maximum in the Fraunhofer diffraction pattern of a slit of width 12×10^{-5} cm when the slit is illuminated by monochromatic light of wavelength 6000 Å.

(1) 30°

(2) 45°

(3) 60°

(4) 15°

7) White light is used to illuminate two slits in a Young's double slit experiment. The separation between the slits is b and the screen is at a distance d(>>b) from the slits. At a point on the screen directly in front of one of the slits, certain wavelengths are missing Some of these missing wavelengths are :-

(i) $\lambda = \frac{b^2}{d}$ (ii) $\lambda = \frac{2b^2}{d}$

(iii) $\lambda = \frac{b^2}{3d}$ (iv) $\lambda = \frac{2b^2}{3d}$

(1) (i), (ii)

(2) (ii), (iii)

(3) (i), (iii)

(4) (ii), (iv)

8) A thin mica sheet of thickness 2×10^{-6} m and refractive index ($\mu = 1.5$) is introduced in the path of the upper wave. The wavelength of the wave used is 5000Å. The central bright maximum will shift :

(1) 2 fringes upward

(2) 2 fringes downward

(3) 10 fringes upward

(4) None of these

9) In Young's double slit experiment the y-co-ordinates of central maximum and 10th maxima are 2 cm and 5 cm respectively. When the YDSE apparatus is immersed in a liquid of refractive index 1.5, the corresponding y-co-ordinates will be :

(1) 2 cm, 7.5 cm

(2) 3 cm, 6 cm

(3) 2 cm, 4 cm

(4) $\frac{4}{3}$cm, $\frac{10}{3}$cm

10) Young's Experiment is performed in air and fringe width is found to be β. Distance of slits from screen is 1.5D and separation between the slits is d. Same experiment is performed in a liquid of refractive index 1.5. To keep fringe width unchanged required changes are:

(1) New distance of screen from slits is 2.25 D and separation between slits is d.

(2) New distance of screen from slits is D and separation between slits is d

(3) New distance of screen from slits is D and separation between slits is 1.5d

(4) New distance of screen from slits is 2.25D and separation between slits is $\frac{2d}{3}$

11) If the ratio of the intensity of two coherent sources is 4 then the fringe visibility of the fringes is :

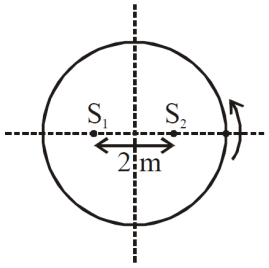
(1) 4

(2) $\frac{4}{5}$

(3) $\frac{3}{5}$

(4) 9

12) Two point sources separated by 2.0 m are radiating in phase with λ = 0.50 m. A detector moves in a circular path around the two sources in a plane containing them. How many maxima are



detected?

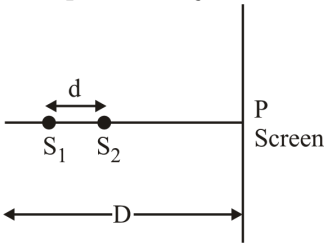
(1) 16

(2) 20

(3) 24

(4) 32

13) Two coherent point sources S_1 and S_2 are separated by a small distance d as shown in the figure.



The fringes obtained on the screen will be

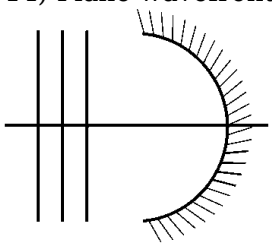
(1) semi – circles

(2) concentric circles

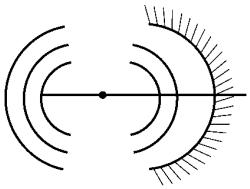
(3) points

(4) straight lines

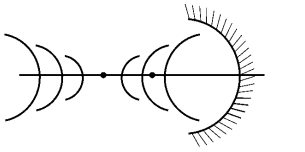
14) Plane wavefronts are incident on a spherical mirror as shown. The reflected wavefronts will be :-



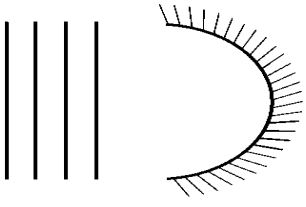
(1)



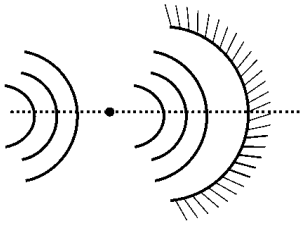
(2)



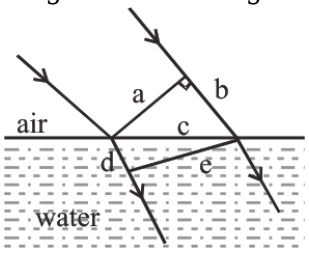
(3)



(4)



15) Figure shows plane wave refracted from air to water using Huygen's principle a, b, c, d, e are lengths on the diagram. The refractive index of water w.r.t. air is in the ratio :–



(1) a/e

(2) b/e

(3) b/d

(4) d/b

16) In a double slit experiment, instead of taking slits of equal widths, one slit is made twice as wide as the other. Then, in the interference pattern.

(1) the intensities of both the maxima and the minima increase

(2) the intensity of the maxima increases and the minima has zero intensity

(3) the intensity of the maxima decreases and that of the minima increases

(4) the intensity of the maxima decreases and the minima has zero intensity.

17) In a young's double slit experiment the intensity at a point where the path difference is $\frac{\lambda}{6}$ (λ being the wavelength of light used) is I. If I^0 denotes the maximum intensity then $\frac{I}{I_0}$ is equal to :–

(1) $\frac{1}{\sqrt{2}}$

(2) $\frac{\sqrt{3}}{2}$

(3) $\frac{1}{2}$

(4) $\frac{3}{4}$

18) In YDSE the slits separated by d = 0.8 mm are illuminated by λ = 7200 Å. Interference pattern is obtainend on a screen 2m from the slits. Find minimum distance from central bright at which the intensity is 50% of the maximum?

(1) 0.45×10^{-3} m

(2) 0.45×10^{-6} m

(3) 0.70×10^{-3} m

(4) 0.70×10^{-6} m

19) In an interference pattern, at a point we observe the 16th order bright fringe or maxima for λ_1=7200Å. What order will be visible here, if the source is replaced by light of wavelength λ_2=4800Å.

(1) 12

(2) 24

(3) 36

(4) 16

20) In young's double slit experiment the amplitude of two sources are 3a and a respectively. The ratio of intensities of bright and dark fringe will be -

(1) 3 : 1

(2) 9 : 1

(3) 2 : 1

(4) 4 : 1

21) Which of the following motion is oscillatory –

(1) $a = -4(x - 2)^3$

(2) $a = 2(x - 2)^6$

(3) $a = +10(x - 3)^3$

(4) $a = 10(x - 3)^2$

22) A particle excutes SHM on a straight line path. The amplitude of oscillation is 2 cm. When the displacement of the particle from the mean position is 1 cm, the numerical value of magnitude of acceleration, is equal to the numerical value of magnitude of velocity. The frequency of SHM (in second^{-1}) is :-

(1) $2\pi\sqrt{3}$

(2) $\frac{2\pi}{\sqrt{3}}$

(3) $\frac{\sqrt{3}}{2\pi}$

(4) $\frac{1}{2\pi\sqrt{3}}$

23) A particle moves with simple harmonic motion in a straight line. In first τ s, after starting from rest it travels a distance a, and in next τ s it travels 2a, in same direction, then :

(1) amplitude of motion is 3a

(2) time period of oscillations is 8τ

(3) amplitude of motion is 4a

(4) time period of oscillations is 6τ

24) A particle executes simple harmonic motion with amplitude A and time period T. The minimum time taken by the particle to reach $x = \frac{\sqrt{3}}{2}A$ from $x = \frac{A}{\sqrt{2}}$ will be :

(1) $\frac{T}{8}$

(2) $\frac{T}{6}$

(3) $\frac{T}{12}$

(4) $\frac{T}{24}$

25) A particle executes simple hamonic motion with a time period of 16s. At time t = 2s, the particle crosses the mean position while at t = 4s, its velocity is 4 ms^{-1}. The amplitude of motion (in metre) is :-

(1) $\sqrt{2}\pi$

(2) $\frac{16\sqrt{2}}{\pi}$

(3) $24\sqrt{2}\pi$

(4) $\frac{32\sqrt{2}}{\pi}$

26) A man of mass 60 kg standing on a platform executing S.H.M. in the vertical plane. The displacement from the mean position varies as
y = 0.5 sin (2π ft). The value of f, for which the man will feel weightlessness at the highest point is:

(y is in metres)

(1) $\frac{g}{4\pi}$

(2) 4 πg

(3) $\frac{\sqrt{2g}}{2\pi}$

(4) $2\pi\sqrt{2g}$

27) For a particle executing SHM along a straight line, match the statements in column-I with statement in column-II. (Note that displacement given in column-I is to be measured from mean position).

	Column-I		**Column-II**
(A)	Velocity-displacement graph may be	(p)	straight line
(B)	Acceleration-velocity graph may be	(q)	circle
(C)	Acceleration-displacement graph will be	(r)	ellipse
(D)	Acceleration-time graph will be	(s)	sinusoidal curve

(1) (A) q,r (B) q,r (C) p (D) s

(2) (A) s (B) q,r (C) p (D) q,r

(3) (A) p (B) q,r (C) q,r (D) s

(4) (A) p (B) s (C) q,r (D) q,r

28) What fraction of the total energy is potential, when the displacement is one-half of the amplitude ?

(1) 1/4

(2) 2/4

(3) 3/4

(4) 3.5/4

29) **Assertion (A) :-** The graph of total mechanical energy of a particle in SHM with respect to position is a straight line with zero slope.
Reason (R) :- Total energy of particle in SHM remains constant throughout its motion.

(1) Both **(A)** and **(R)** are true and **(R)** is the correct explanation of **(A)**.

(2) Both **(A)** and **(R)** are true and **(R)** is NOT the correct explanation of **(A)**.

(3) **(A)** is true but **(R)** is false.

(4) **(A)** is false but **(R)** is true.

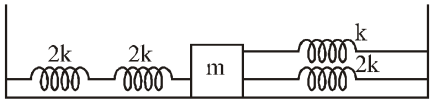
30) PE of a body of mass m is $U = a + \frac{bx^2}{2}$.
Find its time period of SHM ?

(1) $\frac{2\pi}{b}$

(2) $\frac{2\pi}{\sqrt{b}}$

(3) $2\pi\sqrt{\frac{b}{m}}$

(4) $2\pi\sqrt{\frac{m}{b}}$

31) 

The frequency of oscillation of the system shown in figure is :-

(1) $\frac{1}{2\pi}\sqrt{\frac{k}{4m}}$

(2) $\frac{1}{2\pi}\sqrt{\frac{4k}{m}}$

(3) $\frac{1}{2\pi}\sqrt{\frac{k}{7m}}$

(4) $\frac{1}{2\pi}\sqrt{\frac{7k}{m}}$

32) A clock S is based on oscillation of a spring and a clock P is based on pendulum motion. Both clocks run at the same rate on earth. On a planet having the same density as earth, but twice the radius :

(1) S will run faster than P

(2) P will run faster than S

(3) both run at the same rate as on the earth

(4) both run at equal rates but not the same as on the earth

33) Two bodies P and Q of equal masses are suspended from two separate massless springs of force constants k_1 and k_2 respectively. If the two bodies oscillate vertically such that their maximum velocities are equal, then ratio of the amplitude of P to that of Q is:

(1) $\sqrt{\frac{k_1}{k_2}}$

(2) $\frac{k_1}{k_2}$

(3) $\sqrt{\frac{k_2}{k_1}}$

(4) $\frac{k_2}{k_1}$

34) A simple pendulum of length L and mass (bob) M is oscillating in a plane about a vertical line between angular limits $-\phi$ and $+\phi$. For an angular displacement $\theta(|\theta| < \phi)$, the tension in the string

and the velocity of the bob are T and V respectively. The following relations hold good under the above conditions :

(i) $T\cos\theta = mg$

(ii) $T - Mg\cos\theta = \frac{Mv^2}{L}$

(iii) The magnitude of the tangential acceleration of the bob $[a_T] = g\sin\theta$

(iv) $T = Mg\cos\theta$

(1) (i) and (iii)

(2) (ii) and (iii)

(3) (i),(ii) and (iii)

(4) only (iv)

35) A pendulum suspended from the ceiling of an elevator at rest has a time period T_1. When the elevator moves up with an acceleration a, its time period becomes T_2. When the elevator moves down with an acceleration a, its time period becomes T_3, then :-

(1) $T_1 = \sqrt{(T_2T_3)}$

(2) $T_1 = \sqrt{(T_2^2 + T_3^2)}$

(3) $T_1 = \frac{T_2T_3\sqrt{2}}{\sqrt{(T_2^2 + T_3^2)}}$

(4) None of these

36) The time period of a seconds' pendulum is 2s. The spherical bob which is empty from inside has a mass of 50 g. This is now replaced by another solid bob of the same radius but having a different mass of 100 g. The new time period will be :-

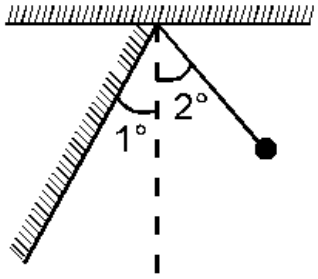
(1) 4 s

(2) 1 s

(3) 2 s

(4) 8 s

37) A simple pendulum of length 1 m is allowed to oscillate with amplitude 2°. It collides elastically with a wall inclined at 1° to the vertical. Its time period will be (use $g = \pi^2$) :-

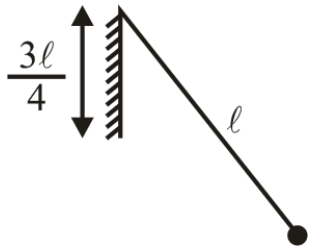


(1) 2/3 sec

(2) 4/3 sec

(3) 2 sec

(4) none of these



38) Find time period of oscillation of the given pendulum :-

(1) $\frac{3\pi}{2}\sqrt{\frac{\ell}{g}}$

(2) $\pi\sqrt{\frac{\ell}{g}}$

(3) $2\pi\sqrt{\frac{\ell}{g}}$

(4) $\frac{2\pi}{3}\sqrt{\frac{\ell}{g}}$

39) The displacement of a particle varies according to the relation
$x = 3 \sin 100t + 8 \cos^2 50t$. Which of the following is INCORRECT about this motion ?

(1) The motion of the particle is S.H.M.

(2) The amplitude of the S.H.M. of the particle is 5 units

(3) The amplitude of the resultant S.H. M. is $\sqrt{73}$ units

(4) The maximum displacement of the particle from the origin is 9 units

40) The mass and the diameter of a planet are three times the respective values for the earth. The time period of oscillation of a simple pendulum on earth is 2s. The time period of oscillation of the same pendulum on the planet would be :

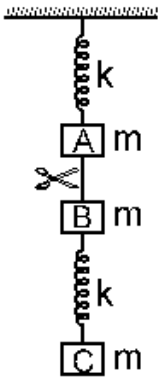
(1) $2\sqrt{3}s$

(2) $\frac{\sqrt{3}s}{2}$

(3) $\frac{2s}{\sqrt{3}}$

(4) $\frac{3s}{2}$

41) The spring block system as shown in figure is in equilibrium. The string connecting blocks A and B is cut. The mass of all the three blocks is m each and spring constant of each spring is k. The



amplitude of resulting oscillation of block A is

(1) $\frac{mg}{k}$

(2) $\frac{2mg}{k}$

(3) $\frac{3mg}{k}$

(4) $\frac{4mg}{k}$

42) **Assertion :** Water in a U-tube executes SHM. The time period for mercury filled up to the same height in the U-tube will be greater than that in case of water.
Reason : The amplitude of an oscillating pendulum goes on increasing.

(1) Both Assertion and Reason are true and Reason is the correct explanation of Assertion.

(2) Both Assertion and Reason are true but Reason is not the correct explanation of Assertion.

(3) Assertion is true but Reason is false.

(4) Both Assertion and Reason are false.

43) Two particles execute simple harmonic motion of the same amplitude and frequency along the same straight line. They pass one another when going in opposite directions and each time their displacement is half the amplitude. The phase difference between them is :–

(1) 45°

(2) 60°

(3) 90°

(4) 120°

44) Two particles P and Q, describe SHM of same amplitude 'a' and frequency υ along the same straight line. The maximum distance between two particles is $\sqrt{2}a$. The initial phase difference between the particles is :-

(1) zero

(2) π/2

(3) π/6

(4) π/3

45) A spring of spring constant K is cut into n equal parts, out of which r parts are placed in parallel and connected with mass m. The time period of oscillatory motion of mass m is :

(1) $T = 2\pi\sqrt{\frac{m}{nrK}}$

(2) $T = 2\pi\sqrt{\frac{nrm}{K}}$

(3) $T = 2\pi\sqrt{\frac{rm}{nK}}$

(4) $T = 2\pi\sqrt{\frac{nm}{rK}}$

46) A man fires a bullet standing between two cliffs. First echo is heard after 3 seconds and second echo is heard after 5 seconds. If the velocity of sound is 330 m/s, then the distance between the cliffs is :

(1) 1650 m

(2) 1320 m

(3) 990 m

(4) 660 m

47) The first two resonance lengths in a resonance tube formed are 16.5 cm and 51 cm. The end correction for the tube is :-

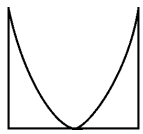
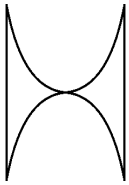
(1) 0.25 cm

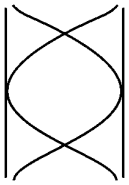
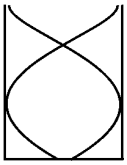
(2) 0.50 cm

(3) 0.75 cm

(4) 1.00 cm

48) The vibration of four air columns under identical conditions are represented in the figure below. The ratio of frequencies $n_p : n_q : n_r : n_s$ will be:
(Lenths are same) :-

(p)  (q) 

(r)  (s) 

(1) 12 : 6 : 3 : 4

(2) 1 : 2 : 4 : 3

(3) 4 : 2 : 3 : 1

(4) 6 : 2 : 3 : 4

49) An open pipe produces a fundametal frequency *f* in air. When dipped in water, so that 2/5 of its length is under water, its fundamental frequency is :-

(1) $\frac{12}{5}f$

(2) $\frac{6}{5}f$

(3) $\frac{5}{12}f$

(4) $\frac{5}{6}f$

50) Following are expressions for four plane simple harmonic waves :

(a) $y_1 = A\cos 2\pi\left(n_1 t + \frac{x}{\lambda_1}\right)$

(b) $y_2 = A\cos 2\pi\left(n_1 t + \frac{x}{\lambda_1} + \pi\right)$

(c) $y_3 = A\cos 2\pi\left(n_2 t + \frac{x}{\lambda_2}\right)$

(d) $y_4 = A\cos 2\pi\left(n_2 t - \frac{x}{\lambda_2}\right)$

The pairs of waves which will produce destructive interference and stationary waves respectively in a medium, are :

(1) (c, d), (a, b)

(2) (a, d), (b, c)

(3) (a, c), (b, d)

(4) (a, b), (c, d)

51) A source of sound gives five beats per second when sounded with another source of frequency 100 s^{-1}. The second harmonic of the source together with a source of frequency 205 s^{-1} gives five beats per second. What is the frequency of the source :-

(1) 105 s^{-1}

(2) 205 s^{-1}

(3) 95 s^{-1}

(4) 100 s^{-1}

52) A tuning fork of frequency 512 Hz makes 4 beats per second with the vibrating string of a piano. The beat frequency decreases to 2 beats per second when the tension in the piano string is slightly increased. The frequency of the piano string before increasing the tension was :-

(1) 510 Hz

(2) 514 Hz

(3) 516 Hz

(4) 508 Hz

53) Two columns of some statements are given, find correct match for them :

	Column-I		**Column-II**

(a)	Path diff. created for successive max. intensity	(i)	λ/2
(b)	Path diff. created for intensity to become min. to max.	(ii)	λ
(c)	Path diff. for destructive interference	(iii)	2λ
(d)	Path diff. for constructive interference	(iv)	5λ/2

(1) a-(ii); b-(iii); c-(iv); d-(i)

(2) a-(ii); b-(iv); c-(i) (iv); d-(ii) (iii)

(3) a-(ii); b-(i); c-(i) (iv); d-(ii) (iii)

(4) a-(i); b-(iv); c-(iv); d-(i) (iv)

54) A sound has an intensity of 2×10^{-8} W/m^2 . Its intensity level (in decibel) is ($\log_{10} 2 = 0.3$) :-

(1) 23

(2) 3

(3) 43

(4) 4.3

55) ***Assertion :-*** The speed of sound in solids is maximum though their density is large.
Reason :- The coefficient of elasticity of solid is very large.

(1) Both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.

(2) Both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.

(3) Assertion is True but the Reason is False.

(4) Both Assertion & Reason are False.

56) The ratio of the velocity of sound in hydrogen ($\gamma = 7/5$) to that in helium ($\gamma = 5/3$) at the same temperature is :

(1) $\sqrt{\frac{5}{42}}$

(2) $\sqrt{\frac{5}{21}}$

(3) $\sqrt{\frac{42}{25}}$

(4) $\sqrt{\frac{21}{5}}$

57) The length of a sonometer wire is 1.25 m and density is 8×10^3 kg/m^3. It can bear a stress of 3.2×10^8 N/m^2 without exceeding the elastic limit. The fundamental frequency that can be produced in

the wire, is :-

(1) 100 Hz
(2) 80 Hz
(3) 200 Hz
(4) 250 Hz

58) Two stretched strings of same material are vibrating under same tension in fundamental mode. The ratio of their frequencies is 1 : 2 and ratio of vibrating length is 1 : 4. Then the ratio of radii of the string is :

(1) 2 : 1
(2) 4 : 1
(3) 3 : 2
(4) 8 : 1

59) A sitar wire is replaced by another wire of same length and material, but of triple radius. If the tension in the wire remains the same, the frequency will become :-

(1) nine times
(2) three times
(3) one-third
(4) one-ninth

60) A second harmonic has to be generated in a string of length stretched between two rigid supports. The points where the string has to be plucked and touched are –

(1) Pluck at $\frac{\ell}{2}$, touch at $\frac{3\ell}{4}$
(2) Pluck at $\frac{\ell}{2}$, touch at $\frac{\ell}{4}$
(3) Pluck at $\frac{\ell}{4}$, touch at $\frac{3\ell}{4}$
(4) Pluck at $\frac{\ell}{4}$, touch at $\frac{\ell}{2}$

61) The equation of a plane progressive wave is

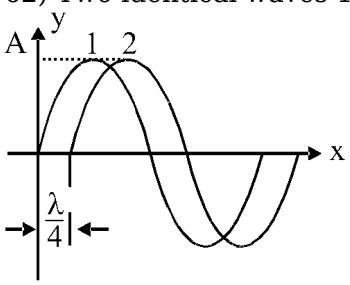
$y = 0.09 \sin 8\pi\left(t - \frac{x}{20}\right)$

When it is reflected at rigid support, its amplitude becomes (2/3)rd of its previous value. The equation of the reflected wave is :-

(1) $y = 0.09 \sin 8\pi\left(t - \frac{x}{20}\right)$
(2) $y = 0.06 \sin 8\pi\left(t - \frac{x}{20}\right)$
(3) $y = 0.06 \sin 8\pi\left(t + \frac{x}{20}\right)$

(4) $y = -0.06 \sin 8\pi \left(t + \frac{x}{20}\right)$

62) Two identical waves 1 and 2 each of intensity I_0 are superimposed. The resulting intensity is :



(1) I_0

(2) $2\sqrt{2}I_0$

(3) $4I_0$

(4) $2I_0$

63) Two mechanical waves, $y_1 = 2 \sin 2\pi (50\ t - 2x)$ & $y_2 = 4 \sin 2\pi (ax + 100\ t)$ propagate in a medium with same speed.

(1) The ratio of their intensities is 1: 16

(2) The ratio of their intensities is 1: 4

(3) The value of 'a' is 8 units

(4) The value of 'a' is 2 units

64) If the equation of transverse wave is

$y = 5 \sin 2\pi \left(\frac{t}{0.04} - \frac{x}{40}\right)$

where distance is in cm and time in second, then the wavelength of wave will be :-

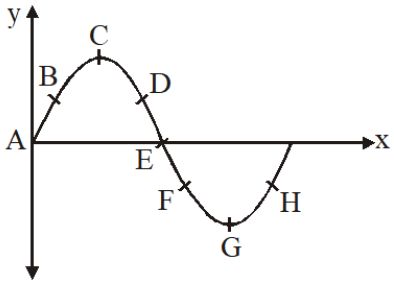
(1) 20 cm

(2) 40 cm

(3) 60 cm

(4) None of these

65) A transverse wave is travelling along a string from left to right. The adjoining figure represents the shape of the string at a given instant. At this instant, among the following , choose the wrong statement:-



(1) Points D, E and F have upwards positive velocity.

(2) Points A, B and H have downwards negative velocity.

(3) Points C and G have zero velocity.

(4) Points A and E have minimum velocity.

66) If wavelength of a wave is λ = 6000Å. Then wave number will be :-

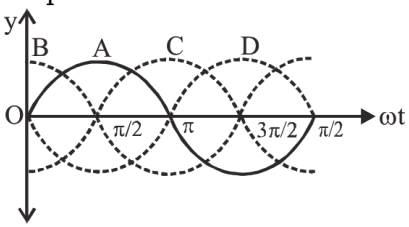
(1) $166 \times 10^{3}\ m^{-1}$

(2) $16.6 \times 10^{-1}\ m^{-1}$

(3) $1.66 \times 10^{6}\ m^{-1}$

(4) $1.66 \times 10^{7}\ m^{-1}$

67) Figure given below shows four progressive waves A, B, C and D with their phases expressed with respect to the wave A. It can be calculated from the figure that :-



(1) the wave C is ahead by a phase angle of π/2 and the wave B lags behind by a phase angle of π/2

(2) the wave C lags behind by a phase angle of π/2 and the wave B is ahead by a phase angle of π/2

(3) the wave C is ahead by a phase angle of π and the wave B lags behind by a phase angle of π.

(4) the wave C lags behind by a phase angle of π and the wave B is ahead by a phase angle of π.

68) The linear mass density of a vibrating string is 1.3×10^{-4} kg/m. A transverse wave is propagating on the string and is described by the equation y = 0.021 sin (x+30t) where x and y are measured in meter and t in second the tension in the string is approximately:–

(1) 0.12 N

(2) 0.48 N

(3) 1.20 N

(4) 4.80 N

69) A wire of linear mass density 9×10^{-3} kg/m is stretched between two rigid supports under a tension of 360 N. The wire resonates at frequency 210 Hz. The next higher frequency at which the same wire resonates is 280 Hz. The number of loops produced in the first case will be :-

(1) 1

(2) 2

(3) 3

(4) 4

70) Two tuning forks have frequency 256 Hz (A) and 262 Hz (B). A produces some beats per second with unknown tuning fork, same unknown tuning fork produce double beats per second from B tuning fork then the frequency of unknown tuning fork is (in Hz) :-

(1) 262

(2) 260

(3) 250

(4) 300

71) An organ pipe P_1 closed at one end vibrating in its first overtone and another pipe P_2 open at both ends vibrating in its third harmonic are in resonance with a given tuning fork. The ratio of the length of P_1 to that of P_2 is :-

(1) 8/3

(2) 1/6

(3) 1/2

(4) 1/3

72) ***Assertion :-*** Velocity of particles, while crossing mean position in a stationary wave, varies from maximum at antinodes to zero at nodes.
Reason :- Amplitude of vibration at antinodes is maximum and it is zero at nodes and all particles between two successive nodes cross the mean position together.

(1) Both Assertion and Reason are true and reason is a correct explanation of Assertion.

(2) Both Assertion and Reason are true but Reason is not a correct explanation of Assertion.

(3) Assertion is True but Reason is false.

(4) Both Assertion and Reason are false.

73) For a closed organ pipe, match the entries of column I with the entries of coloum II.

	Column-I		**Column-II**
(A)	Third overtone frequency is x times the fundamental frequency. Here x is equal to	(P)	3
(B)	Number of nodes in second overtone	(Q)	4
(C)	Number of antinodes in second overtone	(R)	5
		(S)	7

(1) A → S, B → P, C → P

(2) A → R, B → P, C → Q

(3) A → P, B → R, C → Q

(4) A → R, B → P, C → P

74) A cylindrical tube (L = 120 cm.) is vibrated with a tuning fork of frequency 330 Hz. If it is filling by water to get resonance, minimum length of water column is (v_{air} = 330 m/s) :–

(1) 45 cm

(2) 60 cm

(3) 25 cm

(4) 20 cm

75) In resonance tube first resonating length is 17 cm second resonating length is 55 cm, then radius of tube will be ?

(1) $\frac{2}{3}$ cm

(2) $\frac{10}{3}$ cm

(3) $\frac{3}{10}$ cm

(4) $\frac{3}{2}$cm

ANSWER KEYS

PHYSICS

Q.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
A.	**2**	**1**	**3**	**3**	**1**	**1**	**3**	**1**	**3**	**1**	**2**	**1**	**2**	**1**	**3**	**1**	**4**	**1**	**2**	**4**
Q.	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40
A.	**1**	**3**	**4**	**4**	**4**	**3**	**1**	**1**	**1**	**4**	**2**	**2**	**3**	**2**	**3**	**3**	**2**	**1**	**3**	**1**
Q.	41	42	43	44	45	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60
A.	**2**	**4**	**4**	**2**	**1**	**2**	**3**	**2**	**4**	**4**	**1**	**4**	**3**	**3**	**1**	**3**	**2**	**4**	**3**	**4**
Q.	61	62	63	64	65	66	67	68	69	70	71	72	73	74	75					
A.	**4**	**4**	**1**	**2**	**4**	**3**	**2**	**1**	**3**	**3**	**3**	**1**	**1**	**1**	**2**					

SOLUTIONS

PHYSICS

2) $20\frac{\lambda}{d} = \frac{2\lambda}{a} \Rightarrow a = \frac{d}{10} = 0.2mm$

3)

$i_p = 53°4'$ and $i_p + r = 90°$

4) After first, $I_1 = \frac{I_0}{2}$
After second, $I_2 = I_1 \cos^2 30°$
$= \frac{I_0}{2} \times \frac{3}{4} = \frac{3I_0}{8}$
After third, $I_3 = I_2 \cos^2 30°$
$= \frac{3I_0}{8} \times \frac{3}{4} = \frac{9I_0}{32}$
After fourth, $I_3 \cos^2 30°$
$= \frac{9I_0}{32} \times \frac{3}{4} = \frac{27}{128} I_0 = 0.21\ I_0$

5)

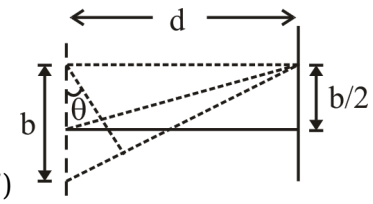
$\sin\theta = \frac{3\lambda}{2a}$
$\Rightarrow \theta = \frac{3\lambda}{2a}$
$\theta = \frac{3 \times 5890 \times 10^{-10}}{2 \times 25 \times 10^{-5}}$
$= 3.534 \times 10^{-3}$ rad.

6)

$\sin\theta = \frac{\lambda}{a}$

θ = half angular width of the central maximum.
$a = 12 \times 10^{-5}$ cm, $\lambda = 6000$ Å $= 6 \times 10^{-5}$ cm

$\sin\theta = \frac{\lambda}{a} = \frac{6 \times 10^{-5}}{12 \times 10^{-5}} = 0.50 \Rightarrow \theta = 30°$



7)

$\sin\theta = \frac{\Delta}{b}$ ...(1)

For small angle $\sin\theta = \frac{b}{2d}$ ...(2)

by (1) & (2)

$\frac{\Delta}{b} = \frac{b}{2d}$

$\Delta = \frac{b^2}{2d}$' for missing wavelengths

$\Delta = (2n-1)\frac{\lambda}{2}$, n = 1, 2, 3, ...

$\Delta = \frac{\lambda}{2}, \frac{3\lambda}{2}, \frac{5\lambda}{2}$ .....

So, $\frac{\lambda}{2} = \frac{b^2}{2d} \Rightarrow \lambda = \frac{b^2}{d}$

$\frac{3\lambda}{2} = \frac{b^2}{2d} \Rightarrow \lambda = \frac{b^2}{3d}$

$\frac{5\lambda}{2} = \frac{b^2}{2d} \Rightarrow \lambda = \frac{b^2}{5d}$ or $\lambda = \frac{b^2}{(2n-1)d}$

8) Shift = $\frac{\beta}{\lambda}(\mu - 1)t$

$= \frac{\beta}{(5000 \times 10^{-10})} \times (1.5 - 1) \times 2 \times 10^{-6}$

$= 2\beta$ i.e., 2 fringes upwards

9)

$10\beta_1 = 5 - 2 = 3$cm in medium

$10\beta_2 = \frac{10\beta_1}{\mu} = \frac{20}{3}\beta_1 = \frac{2}{3}$ (3 cm)

$10\beta_2 = 2$ cm

⇒ coordinates of central maxima remain same 2 cm.

⇒ coordinates of 10th maxima = 2 cm + 10 β_2

= 2 + 2 = 4 cm

10) $\beta = \frac{\lambda(1.5\ D)}{d}$

As β = β'

$= \frac{1.5\lambda D}{d} = \frac{\lambda D'}{1.5\,d}$

D' = 2.25 D

11) $\frac{I_1}{I_2} = 4,\ \text{F.V.} = \frac{I_{max} - I_{min}}{I_{max} + I_{min}}$

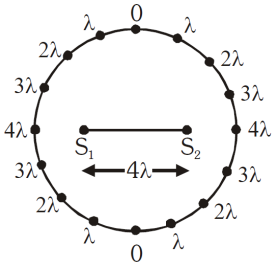
$= \frac{2\sqrt{I_1 I_2}}{I_1 + I_2} = \frac{2 \times 2}{5} = 4/5$

12) $\lambda = 0.5$ m

d = 2m

So distance between S_1 & S_2 $(0.5\ \lambda) \times 8$

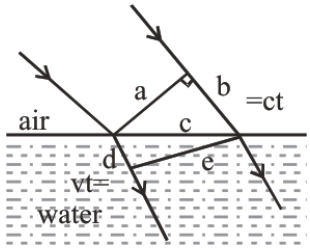
$= 4\ \lambda$



16 maxima

13) Path difference on circles around 'P' is same. So, the fringes obtained on the screen in the given condition will be concentric circles.

15)



Time taken by wave in air to travel distance

$t = \frac{b}{c}$

During same time distance travelled in medium.

$d = tv \qquad \text{so} \qquad \mu = \frac{c}{v} = \frac{b/t}{d/t} = \frac{b}{d}$

17)

$\Delta = \frac{\lambda}{6}$

$\therefore \phi = \frac{2\pi}{\lambda} \times \frac{\lambda}{6} = \frac{\pi}{3}$

$\therefore I = I^0 \cos^2\left(\frac{\pi}{6}\right)$

$= I^0 \times \left(\frac{\sqrt{3}}{2}\right)$

$I = I^0 \times \frac{3}{4}$

$\therefore \frac{I}{I_0} = \frac{3}{4}$

18) $I = \frac{I_{max}}{2} \Rightarrow \frac{I_{max}}{2} = I_{max}\cos^2\frac{\phi}{2}$

$\cos^2\frac{\phi}{2} = \frac{1}{2} \Rightarrow \phi = \frac{\pi}{2}$

$\Delta x = \frac{\lambda}{4}$

So $y = \Delta x.\frac{D}{d} = \frac{\lambda}{4}.\frac{D}{d} = \frac{7200 \times 10^{-10}}{4} \times \frac{2}{0.8 \times 10^{-3}}$

$= 0.45 \times 10^{-3}$ m

19) The distance of a bright fringe from zero order fringe is given by

$x_n = \frac{n\lambda D}{2d}$

Thus at a given point nλ=constant

$\therefore \quad n_1\lambda_1 = n_2\lambda_2$

$n_2 = \frac{n_1\lambda_1}{\lambda_2} = \frac{16 \times 7200}{4800} = 24$

20) $\frac{I_{max}}{I_{min}} = \frac{(3a + a)^2}{(3a - a)^2} = \frac{16a^2}{4a^2} = \frac{4}{1}$

21) General equation for oscillatory
Motion $a \propto -(x - B)^n$ [a ⇒ acceleration]
where n = odd number

22) velocity, $v = \omega\sqrt{a^2 - (a/2)^2} = \frac{\sqrt{3}}{2}a\omega$

Acceleration$^f = \frac{\omega^2 a}{2}$

$\therefore$ v = f (given)

$\therefore \frac{\omega^2 a}{2} = \frac{\sqrt{3}}{2}a\omega$

$2\pi n = \sqrt{3}$

$n = \frac{\sqrt{3}}{2\pi}$

23)

$A - a = A\cos\omega t$

$A - (a + 2a) = A\cos 2\omega t$

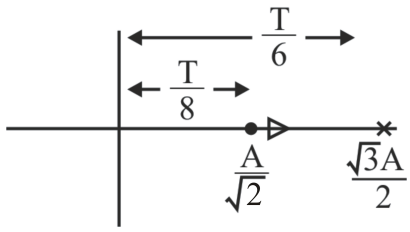
$\cos\omega t = \frac{A - a}{A}$

$\cos 2\omega t = \frac{A-3a}{A}$

Solving we get A = 2a

& $\frac{\pi}{3} = \omega\tau \Rightarrow T = 6\tau$

24)



Minimum time taken to reach

$\frac{A}{\sqrt{2}}$ to $\frac{\sqrt{3}A}{2}$

$= \frac{T}{6} - \frac{T}{8}$

$= \frac{T}{24}$

25)

At t = 2s,

$0 = A\sin(\omega t + \phi)$

$\Rightarrow \phi = -\omega t = \frac{-2\pi}{16} \times 2 = \frac{-\pi}{4}$

At t = 4s,

$v = A\omega \cos\left(\omega t - \frac{\pi}{4}\right) \Rightarrow 4 = A\left(\frac{\pi}{8}\right)\cos\frac{\pi}{4}$

$\Rightarrow A = \frac{32\sqrt{2}}{\pi}$

26)

$\omega^2 A = g$

$4\pi^2 f^2 \times \frac{1}{2} = g$

$\Rightarrow f^2 = \frac{2g}{4\pi^2}$

27)

(A) $v = \omega\sqrt{A^2 - x^2}$

$V^2 + \omega^2 x^2 = \omega^2 A^2$

(B) $V = \omega A \cos(\omega t)$

$a = -\omega^2 A \sin(\omega t)$

$\because \sin^2\omega t + \cos^2\omega t = 1$

$\Rightarrow \frac{a^2}{\omega^2 A^2} + \frac{V^2}{\omega^2 A^2} = 1$

(C) $a = -\omega^2 x$

(D) $a = -\omega^2 A \sin(\omega t + \phi)$

28) $TE = \frac{1}{2}kA^2$

$U_x = \frac{1}{2}kx^2$

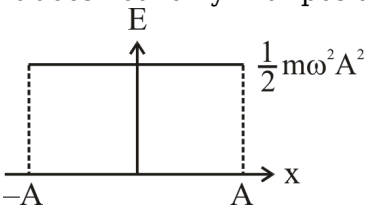
$= \frac{1}{2}k\left(\frac{A^2}{4}\right) = \frac{TE}{4}$

29)

Total energy

$E = \frac{1}{2}m\omega^2A^2$

It does not varry with position i.e. x



30) $F = \frac{-du}{dx} = -bx$

$\Rightarrow a = \frac{-b}{m}x$

$\Rightarrow \omega = \sqrt{\frac{b}{m}} \Rightarrow T = \frac{2\pi}{\omega}$

31)

$f = \frac{1}{2\pi}\sqrt{\frac{k_{eq}}{m}} : k_{eq} = 4k$

$\therefore f = \frac{1}{2\pi}\sqrt{\frac{4k}{m}}$

32) For clock S

on earth $T = 2\pi\sqrt{\frac{m}{k}}$

on planet $T' = 2\pi\sqrt{\frac{m}{k}}$

$T = T'$

For clock P

$T = 2\pi\sqrt{\frac{L}{g}}$

$T' = 2\pi\sqrt{\frac{L}{g'}}$

$$g = \frac{GM}{R^2} = \frac{G \times \frac{4}{3}\pi R^3 \rho}{R^2}$$

$$g = \frac{4}{3}\pi G R \rho$$

$g \propto R$

$g' = 2g$

$$T' = 2\pi\sqrt{\frac{L}{2g}}$$

$T' < T$

so clock P runs faster then S on other planet.

33) $V_1 = V_2$

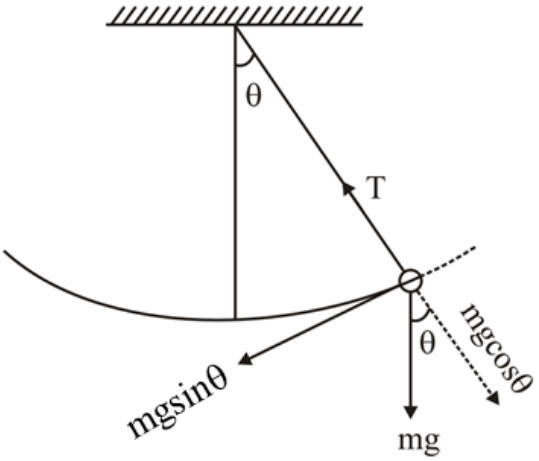
$\omega_1 A_1 = \omega_2 A_2$

$$\frac{A_1}{A_2} = \frac{\omega_2}{\omega_1} = \frac{\sqrt{K_2/m}}{\sqrt{K_1/m}} = \sqrt{\frac{K_2}{K_1}}$$

34) from following figure it is clear that

T – mg cosθ = centripetal force

$$T - mg\cos\theta = \frac{Mv^2}{L}$$



Also tangential force $F_T = mg\sin\theta$

$ma_T = mg\sin\theta$

$\Rightarrow a_T = g\sin\theta$

35) $$T_1 = 2\pi\sqrt{\frac{L}{g}} \quad ...(1)$$

$$T_2 = 2\pi\sqrt{\frac{L}{g+a}} ....(2)$$

$$T_3 = 2\pi\sqrt{\frac{L}{g-a}} ....(3)$$

From (2) $g + a = \frac{4\pi^2 L}{T_2^2}$ ...(4)

From (3) $g - a = \frac{4\pi^2 L}{T_3^2}$ ...(5)

From (1) $g = \frac{4\pi^2 L}{T_1^2}$

equation $\frac{(4) + (5)}{2}$

$$g = \frac{1}{2}4\pi^2 L\left(\frac{1}{T_2^2} + \frac{1}{T_3^2}\right)$$

$$g = \frac{1}{2}gT_1^2\left(\frac{1}{T_2^2} + \frac{1}{T_3^2}\right)$$

$$1 = \frac{T_1^2}{2}\left(\frac{T_2^2 + T_3^2}{T_2^2 \cdot T_3^2}\right) \Rightarrow T_1 = \frac{T_2 T_3\sqrt{2}}{\sqrt{\left(T_2^2 + T_3^2\right)}}$$

Option (3)

36) $T = 2\pi\sqrt{\frac{\ell}{g}}$

 T is independent of mass of bob.

37) Time period for half part :

T = $2\pi\sqrt{\frac{\ell}{g}} = 2\pi\sqrt{\frac{1}{g}} = \frac{2\pi}{\pi} = 2\ \text{sec}.$

So 2° part will be covered in a time

$t = \frac{T}{2} = 1\ \text{sec}.$

For the left 1° part :

$\theta = \theta_0 \sin(\omega t)$

1° = $2^\circ \sin\left(\frac{2\pi}{T} \times t\right) \Rightarrow \frac{1}{2} = \sin\left(\frac{2\pi}{2} \times t\right)$

$\Rightarrow \frac{\pi}{6} = \pi \times t$ $\Rightarrow t = 1/6$ sec.

Total time $\Rightarrow \frac{T}{2} + 2t$

$\Rightarrow$ $1 + 2 \times \frac{1}{6} = 1 + \frac{1}{3} = \frac{4}{3}$ sec.

38) $T = \pi\sqrt{\frac{\ell}{g}} + \pi\sqrt{\frac{\ell}{4g}}$

39)

$x = 3 \sin 100t + 4 \cos 100 t + 4$

$A = 5$

40) $T_e = 2\pi\sqrt{\frac{\ell}{g_e}}$

At other planet $g_p = \frac{GM_p}{R_p^2} = \frac{G(3M_e)}{(3R_e)^2}$

$g_p = \frac{g_e}{3} \Rightarrow T_p = \sqrt{3}T_e = 2\sqrt{3}s$

41) $x_2 = \frac{3mg}{k}$

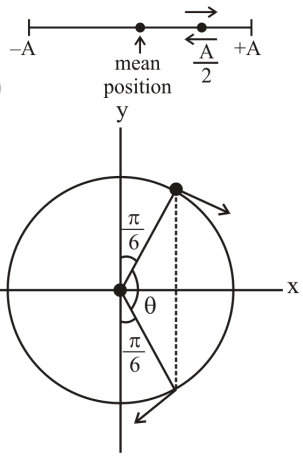
$x_1 = \frac{mg}{k}$

$\therefore A = x_2 - x_1$

$= \frac{2mg}{k}$

42) The period of liquid executing SHM in a U-tube does not depends upon the density of the liquid. Therefore, time period will be the same, when mercury is filled up to the same height as the water in the U-tube.
Now, as the pendulum oscillates, it drags air along with it. Therefore, it's kinetic energy is dissipated in overcoming viscous drag due to air and hence, it's amplitude goes on decreasing.

43)



Phase difference

$\theta = \pi - 2 \times \frac{\pi}{6}$

$= \frac{2\pi}{3} = 120^\circ$

44)

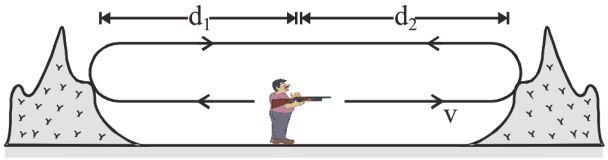


$\phi_1 = \frac{\pi}{4}$

$\phi_2 = \frac{7\pi}{4}$

$\therefore \Delta\phi = \frac{3\pi}{2} \text{or} \frac{\pi}{2}$

45) $T = 2\pi\sqrt{\frac{m}{nrK}}$

46) 

$2(d_1 + d_2) = v(t_1 + t_2)$

$d_1 + d_2 = \frac{330 \times (3+5)}{2} = 1320\,m$

47) $e = \frac{\ell_2 - 3\ell_1}{2} = \frac{51 - 3 \times 16.5}{2} = \frac{51 - 49.5}{2}$

$= \frac{1.5}{2} = 0.75$ cm

48) $n_p = \frac{V}{4\ell}$ ....(1), $n_q = \frac{V}{2\ell}$ .....(2)

$n_r = \frac{2V}{2\ell}$ ...(3), $n_s = \frac{3V}{4\ell}$ .....(4)

$n_p : n_q : n_r : n_s :: \frac{V}{4\ell} : \frac{V}{2\ell} : \frac{2V}{2\ell} : \frac{3V}{4\ell}$

or $n_p : n_q : n_r : n_s :: 1 : 2 : 4 : 3$

49) Open organ pipe $f = \frac{V}{2L}$

when 2/5 of length is under water. the length that is outside water is

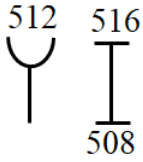
$L' = \frac{3L}{5}$

Fondamental frequency now will be $f' = \frac{V}{4L'} = \frac{V}{4 \times \frac{3L}{5}} = \frac{5}{6} \times \frac{V}{2L} = \frac{5}{6} f$

50)

Phase different between (a) and (b) is π. So distructive interference will take place.
Waves (c) and (d) are travelling in opposite directions, so produce standing wave.

51) Frequency of the source = 100 ± 5
= 105 Hz or 95 Hz
Sound harmonic of the source = 210 Hz or 190 Hz
As the second harmonic gives 5 beats/sec with sound of frequency 205 Hz, the second harmonic should be 210 Hz
⇒ Frequency fo the source = 105 Hz



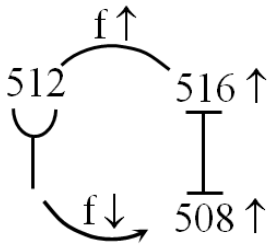
beat frequency

4

52)

Since with increase in Tension frequency of piano string increases

So, after T↑



2

since decreasing condition is satisfied by 508

so Ans 508 Hz

54)

$L = 10 \log_{10}\left(\frac{2 \times 10^{-8}}{10^{-12}}\right)$ decibel

56) $V = \sqrt{\frac{\gamma RT}{M}}$

$$\therefore \frac{V_{H_2}}{V_{He}} = \sqrt{\frac{\gamma_{H_2}}{\gamma_{He}} \cdot \frac{M_{He}}{M_{H_2}}} = \sqrt{\frac{\frac{7}{5}}{\frac{5}{3}} \times \frac{4}{2}} = \sqrt{\frac{42}{25}}$$

57) Velocity of wave,

$$v = \sqrt{\frac{\text{stress}}{\text{density}}} = \sqrt{\frac{3.2 \times 10^8}{8 \times 10^3}}$$

$$= \sqrt{0.4 \times 10^5} = \sqrt{4 \times 10^4} = 200 \text{ m/s}$$

 fundamental frequency $= \frac{v}{2L} = \frac{200}{2 \times 1.25}$

= 80 Hz

58) $\therefore n = \frac{1}{2\ell}\sqrt{\frac{T}{\pi r^2 d}}$

$$\Rightarrow \frac{n_1}{n_2} = \frac{\ell_2}{\ell_1}\sqrt{\frac{r_2^2}{r_1^2}}$$

$$\frac{1}{2} = \frac{4}{1}\sqrt{\frac{r_2^2}{r_1^2}}$$

$$\frac{r_2^2}{r_1^2} = \left(\frac{1}{8}\right)^2$$

$$\boxed{\frac{r_1}{r_2} = 8 : 1}$$

59) For a wire fixed at both ends

$f = \frac{p}{2Lr} \Rightarrow f \propto \frac{1}{r}$

r → triple so f becomes one third

60) Plucking distance

$x = \frac{\ell}{2p} = \frac{\ell}{4}$



touching distance = □/2

61) $A_R = \frac{2}{3} \times A_i = \frac{2}{3} \times 0.09$

$A_R = 0.06$

$y_R = 0.06 \sin\left(8\pi t + \frac{8\pi}{20}x + \pi\right)$

$y_R = -0.06 \sin\left(8\pi t + \frac{8\pi}{20}x\right)$

$y_R = -0.06 \sin\left[8\pi\left(t + \frac{x}{20}\right)\right]$

62) $\phi = \frac{2\pi}{\lambda} \times \frac{\lambda}{4} = \frac{\pi}{2}$

$I = I_1 + I_2 + 2\sqrt{I_1}\sqrt{I_2}\cos\phi$

$= I_0 + I_0 + 2I_0 \cos\frac{\pi}{2} = 2I_0$

63) $I = \frac{P_{avg}}{\tau} = 2\pi^2 A^2 n^2 \rho v \Rightarrow I \propto A^2 n^2$ Same speed $\Rightarrow \frac{50}{2} = \frac{100}{a}$ or a = 4

68)

$T = \mu v^2 = 1.3 \times 10^{-4} \times (30)^2 \simeq 0.12$

69) Ratio of two near by frequency $\frac{210}{280} = \frac{3}{4}$

□ 210 Hz is 3rd harmonic therefore it will have 3 loops.

70)

Let frequency of A, n < 256
⇒ (256 – n) 2 = 262 – n
⇒ n = 512 – 262 = 250 Hz

71) According to question

$$\frac{3v}{4\ell_c} = 3\left(\frac{v}{2\ell_0}\right)$$

$$\Rightarrow \frac{\ell_c}{\ell_0} = \frac{1}{2}$$

73) For p^{th} overtone

$$n = (2P + 1)\frac{v}{4L}$$

No of nodes = No of antinodes = P + 1

74) $\lambda = \frac{v}{f} = \frac{330}{330} = 1m$

L = λ/4, 3λ/4, 5λ/4.... = 25cm, 75cm,125cm
Minimum length of water column =120–75 = 45cm